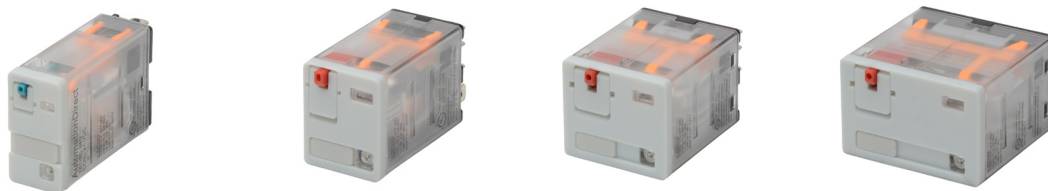


Electromechanical Relays 78 Series

Selection Guide



Electromechanical Relays 78 Series				
Specification	781 Series	782 Series	783 Series	784 Series
Coil Voltages	120 VAC, 240 VAC, 12 VAC, 12 VDC, 24 VAC, 24 VDC	120 VAC, 240 VAC, 12 VAC, 12 VDC, 24 VAC, 24 VDC	120 VAC, 240 VAC, 12 VAC, 12 VDC, 24 VAC, 24 VDC	120 VAC, 240 VAC, 12 VAC, 12 VDC, 24 VAC, 24 VDC
Configuration	SPDT	DPDT	3PDT	4PDT
Contact Rating	15A	15A	15A	15A
Base Socket	5 pin spade terminal	8 pin spade terminal	11 pin spade terminal	14 pin spade terminal
Agency Approvals	UL Recognized (E191059), CE, IEC Std 947-4-1 and 947-5-1, CSA 244610	UL Recognized (E191059), CE, IEC Std 947-4-1 and 947-5-1, CSA 244610	UL Recognized (E191059), CE, IEC Std 947-4-1 and 947-5-1, CSA 244610	UL Recognized (E191059), CE, CSA 244610

Overview

These ice cube style relays are power relays designed for applications demanding high power control in various factory machines and control panels. They are ideal for electrical control panels requiring stable and reliable relays.

Features

- Small package design
- Silver alloy gold flashed contact
- High open contact dielectric strength (up to 2500V rms)
- High reliability and long life
- High vibration and shock resistance
- LED indicator on all models, so you can easily see if the relay is working properly without using a voltmeter
- Flag indicator shows relay status in manual or powered condition
- A pushbutton allows manual operation of the relay without the need for power to the coil
- Lock-Down door, when activated, holds pushbutton and contacts in the "operate" position, allowing circuits to be analyzed.
- SPDT, DPDT, 3PDT and 4PDT models
- Finger grip cover allows easier removal of relays from sockets than conventional relays
- I.D. tag/write labels for identifying relays in multi-relay circuits

Electromechanical Relays 78 Series

Selection Guide

Electromechanical Relays 78 Series							
Part Number	Price	Drawing Link	Coil Voltage	Configuration	Relay Socket Part Number	Price	Drawing Link
781-1C-12D	\$6.00	PDF	12 VDC	SPDT	781-1C-SKT	\$5.00	PDF
781-1C-12A	\$6.00	PDF	12 VAC				
781-1C-24D	\$5.50	PDF	24 VDC				
781-1C-24A	\$6.00	PDF	24 VAC				
781-1C-120A	\$6.00	PDF	120 VAC				
781-1C-240A	\$7.25	PDF	240 VAC				
782-2C-12D	\$7.50	PDF	12 VDC	DPDT	782-2C-SKT	\$5.00	PDF
782-2C-12A	\$7.50	PDF	12 VAC				
782-2C-24D	\$7.25	PDF	24 VDC				
782-2C-24A	\$7.75	N/A	24 VAC				
782-2C-120A	\$7.75	N/A	120 VAC				
782-2C-240A	\$8.75	N/A	240 VAC				
783-3C-12D	\$7.75	PDF	12 VDC	3PDT	783-3C-SKT	\$5.50	PDF
783-3C-12A	\$10.00	PDF	12 VAC				
783-3C-24D	\$10.50	N/A	24 VDC				
783-3C-24A	\$10.50	N/A	24 VAC				
783-3C-120A	\$10.50	N/A	120 VAC				
783-3C-240A	\$10.50	N/A	240 VAC				
784-4C-12D	\$9.75	PDF	12 VDC	4PDT	784-4C-SKT-1	\$5.75	PDF
784-4C-12A	\$12.50	PDF	12 VAC				
784-4C-24D	\$10.00	PDF	24 VDC				
784-4C-24A	\$10.00	N/A	24 VAC				
784-4C-120A	\$10.00	N/A	120 VAC				
784-4C-240A	\$10.00	N/A	240 VAC				

NOTE: Not recommended for low current switching. Find contacts' Minimum Switching Requirement on following page.
For low current switching, please see the QM4N1 and QM4X1 series.

Electromechanical Relays 78 Series Specifications

Electromechanical Relays 78 Series Specifications												
Part Numbers	781-1C-12D	781-1C-12A	781-1C-24D	781-1C-24A	781-1C-120A	781-1C-240A	782-2C-12D	782-2C-12A	782-2C-24D	782-2C-24A	782-2C-120A	782-2C-240A
General Specifications												
*Service Life: Mechanical / Electrical Operations	Mechanical: 10,000,000 operations not powered											
	Electrical: 100,000 operations @ rated resistive load											
Operating Temperature	-40 to +55°C [-40 to +131°F]											
Response Time	20 ms											
Vibration Resistance	± 1mm [10-35 Hz] and 3gn [35-50 Hz]											
Shock Resistance	15 gn											
Weight	26g [0.92 oz]						36g [1.27 oz]					
Environmental Protection	IP40											
NEMA B300 Pilot Duty Rated	Yes											
**Agency Approvals and Standards	UL Recognized File E191059, CE, CSA											
Coil Specifications												
Standard	Mechanical flag indicator, LED Indicator, lockable push to test button											
Coil Input Voltage	12 VDC	12 VAC	24 VDC	24 VAC	120 VAC	240 VAC	12 VDC	12 VAC	24 VDC	24 VAC	120 VAC	240 VAC
Coil Resistance	115Ω	44Ω	450Ω	177Ω	4.43 kΩ	17.72 kΩ	177Ω	44Ω	640Ω	177Ω	4.43 kΩ	17.72 kΩ
Power Consumption	1.4W DC, 1.9W AC @ 50/60 Hz						1.15W DC, 1.4W AC @ 50/60 Hz					
Dropout Voltage (% of nominal voltage or more)	10%	15%	10%	15%			10%	15%	10%	15%		
Pull-in Voltage (% of nominal voltage or less)	85%	85%	85%	85%			80%	85%	80%	85%		
Max. Voltage (Max. continuous voltage)	110% of the rated coil voltage											
Contact Specifications												
Contact Type	SPDT						DPDT					
Contact Material	Silver alloy, gold flashed											
Minimum Switching Requirement	10 mA @ 17 VDC											
Max. Contact Rating	Refer to Contact Ratings charts.											
Dielectric Strength Between Contacts	Between coil contact: 2000V rms; Between poles 2000V rms; Between contacts 1500V rms											

*Note: These devices are rated for 1,000 cycles when used in a motor application. (Per Table 45.1, UL 508).

**Note: UL listed when used with sockets 781-1C-SKT, 782-2C-SKT, 783-3C-SKT, or 784-4C-SKT-1. Current limited to rating of relay or socket, whichever is less.

NEMA Mechanical Switching Ratings and Test Values for AC Control Circuit Contacts											
Contact Rating Designation	Thermal Continuous Test Current (A)	Maximum AC Current, 50/60 Hz (A)								Voltamperes	
		120 Volts		240 Volts		480 Volts		600 Volts			
		Make	Break	Make	Break	Make	Break	Make	Break	Make	Break
B300	5	30	3.00	15	1.50	---	---	---	---	3600	360

This chart is provided as a guideline only, and the ratings and values are not guaranteed to be accurate. It is the users' responsibility to properly size their control circuit devices. The chart values are from NEMA Standard ICS 5-2000, Table 1-4-1.

Contact Ratings 781 Series (current)				
Resistive				*Motor Load
Voltage	Nominal	UL	CSA	UL
28 VDC	15A	15A	12A	---
120 VAC	15A	15A	15A	1/2 Hp
277 VAC	15A	12A	12A	1 Hp

Contact Ratings 782 Series (current)				
Resistive				*Motor Load
Voltage	Nominal	UL	CSA	UL
28 VDC	15A	15A	12A	---
120 VAC	15A	15A	15A	1/2 Hp
277 VAC	15A	12A	12A	1 Hp

Electromechanical Relays 78 Series Specifications

Electromechanical Relay 78 Series Specifications													
Part Numbers	<u>783-3C-12D</u>	<u>783-3C-12A</u>	<u>783-3C-24D</u>	<u>783-3C-24A</u>	<u>783-3C-120A</u>	<u>783-3C-240A</u>	<u>784-4C-12D</u>	<u>784-4C-12A</u>	<u>784-4C-24D</u>	<u>784-4C-24A</u>	<u>784-4C-120A</u>	<u>784-4C-240A</u>	
General Specifications													
*Service Life: Mechanical / Electrical Operations	Mechanical: 10,000,000 operations not powered												
	Electrical: 100,000 operations @ rated resistive load												
Operating Temperature	-40 to +55°C [-40 to +131°F]												
Response Time	20 ms												
Vibration Resistance	± 1mm [10-35 Hz] and 3 gn [35-100 Hz]												
Shock Resistance	15 gn												
Weight	60g [2.12 oz]						80g [2.82 oz]						
Environmental Protection	IP40												
NEMA B300 Pilot Duty Rated	Yes												
**Agency Approvals and Standards	UL Recognized File E191059, CE, CSA												
Coil Specifications													
Standard	Mechanical flag indicator, LED Indicator, lockable push to test button												
Coil Input Voltage	12 VDC	12 VAC	24 VDC	24 VAC	120 VAC	240 VAC	12 VDC	12 VAC	24 VDC	24 VAC	120 VAC	240 VAC	
Coil Resistance	80Ω	30Ω	320Ω	110Ω	2.88 kΩ	11.3 kΩ	76Ω	20Ω	303Ω	80Ω	2.1 kΩ	8kΩ	
Power Consumption	1.85W DC, 2.05W AC @ 50/60 Hz						1.5W DC, 1.5W AC @ 50/60 Hz						
Dropout Voltage (% of nominal voltage or more)	10%	15%	10%	15%			10%	15%	10%	15%			
Pull-in Voltage (% of nominal voltage or less)	80%	85%	80%	85%			80%	85%	80%	85%			
Max. Voltage (Max. continuous voltage)	110% of the rated coil voltage												
Contact Specifications													
Contact Type	3PDT						4PDT						
Contact Material	Silver alloy, gold flashed												
Minimum Switching Requirement	10 mA @ 17 VDC												
Max. Contact Rating	Refer to Contact Ratings charts.												
Dielectric Strength Between Contacts	Between coil and contacts: 2000V rms; Between poles: 2000V rms; Between contacts: 1500V rms												

*Note: These devices are rated for 1,000 cycles when used in a motor application. (Per Table 45.1, UL 508).

**Note: UL listed when used with sockets [781-1C-SKT](#), [782-2C-SKT](#), [783-3C-SKT](#), or [784-4C-SKT-1](#). Current limited to rating of relay or socket, whichever is less.

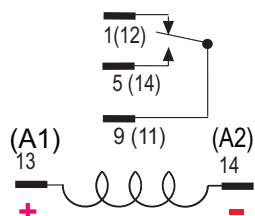
Contact Ratings 783 Series (current)				
Resistive				*Motor Load
Voltage	Nominal	UL	CSA	UL
28 VDC	15A	15A	15A @ 28 VDC 30A max total	–
120 VAC	15A	–	15A	1/2 hp
277 VAC	15A	15A	15A @ 150 VAC 30A max total	1 hp 2 hp max total

Contact Ratings 784 Series (current)				
Resistive				*Motor Load
Voltage	Nominal	UL	CSA	UL
28 VDC	15A	15A	15A @ 28 VDC 30A max total	–
120 VAC	15A	–	15A	1/2 Hp
277 VAC	15A	15A	15A @ 150 VAC 30A max total	1 hp 2 hp max total

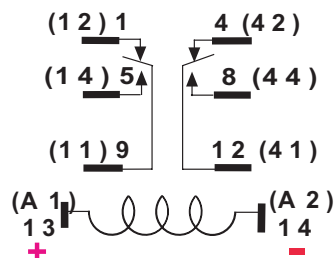
Wiring Diagrams 78 Series

Wiring Diagrams (viewed from pin end)

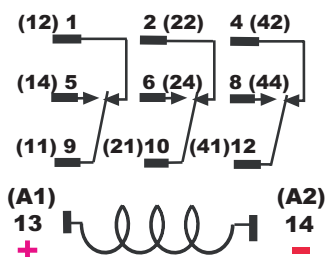
781-1C-XXX



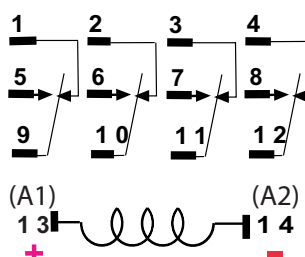
782-2C-XXX



783-3C-XXX

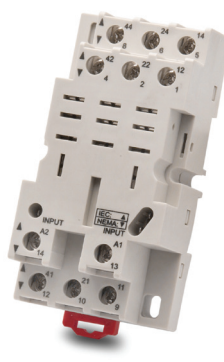


784-4C-XXX



*Note: ALTERNATE NEMA OR IEC () NUMBERS, VIEWED FROM PIN SIDE

Relay Sockets 78 Series

**781-1C-SKT****782-2C-SKT****783-3C-SKT****784-4C-SKT-1**

Relay Sockets 78 Series				
Part Number	Price	Description	Drawing Link	Agency Approval
<u>781-1C-SKT</u>	\$5.00	AutomationDirect relay socket, 35mm DIN rail or panel mount. For use with 781 series cube relays.	<u>PDF</u>	UL Recognized file number: E225080
<u>782-2C-SKT</u>	\$5.00	AutomationDirect relay socket, 35mm DIN rail or panel mount. For use with 782 and AD-70S2 series cube relays.	<u>PDF</u>	
<u>783-3C-SKT</u>	\$5.50	AutomationDirect relay socket, 35mm DIN rail or panel mount. For use with 783 series cube relays.	<u>PDF</u>	
<u>784-4C-SKT-1</u>	\$5.75	AutomationDirect relay socket, 35mm DIN rail or panel mount. For use with 784 series cube relays.	<u>PDF</u>	

Relay Sockets 78 Series Screw Torques and Wire Sizes		
Part Number	Maximum Screw Torques	Maximum Wire Sizes
<u>781-1C-SKT</u>	Terminals 13, 14: 7 in·lbs/0.8 N·m Terminals 1, 5, 9: 9 in·lbs/1.0 N·m	Terminals 13, 14: 18 to 20 AWG, solid or stranded, one or two identical wires Terminals 1, 5, 9: 12 to 20 AWG, solid or stranded, one or two identical wires
<u>782-2C-SKT</u>	All terminals: 9 in·lbs/1.0 N·m	All terminals: 12 to 20 AWG, solid or stranded, one or two identical wires
<u>783-3C-SKT</u>		
<u>784-4C-SKT-1</u>		

Note: Order sockets separately; holding clips are included with sockets.

Packaged M.O.V.s and Diodes

Overview

Metal Oxide Varistors (MOV) and Diode circuits are offered as convenient plug-in modules. Plugging a module into the relay socket connects the circuit in parallel with the relay coil. No additional wiring is required.

Modules fit within the maximum dimensions of the relay and socket.

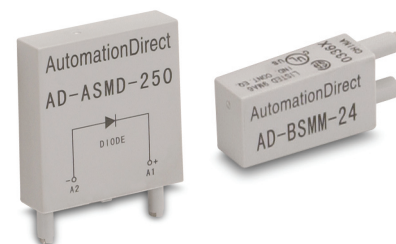
Features

- MOVs protect by shunting potentially damaging electrical spikes away from the relay coil. Ideal for AC and DC applications.
- Diodes protect external drive circuitry from inductive voltages generated when removing coil voltage.
Ideal for DC applications.
Polarity sensitive.

Application

Many PLC systems control one or more inductive load devices. These inductive loads (devices with a coil) generate transient voltages when they are de-energized with a relay contact. When a relay contact is closed it "bounces", which causes the coil to energize and de-energize until the "bouncing" stops. The transient voltage which is generated is much larger in amplitude than the supply voltage, especially with a DC supply voltage.

When switching a DC-supplied inductive load the full supply voltage is always present when the relay contact opens (or "bounces"). When switching an AC-supplied inductive load, if the voltage is not zero when the relay contact opens, there is energy stored in the inductor that is released when the voltage to the inductor is suddenly removed. This release of energy is what produces transient voltages.



When inductive load devices (motors, motor starters, interposing relays, solenoids, valves, etc.) are controlled with relay contacts, it is recommended that a surge suppression device be connected directly across the coil of the field device. If the inductive device has plug-type connectors, the suppression device can be installed on the terminal block of the relay output.

Metal oxide varistors (MOV) and diodes are devices which provide good surge and transient suppression of AC and DC powered coils.

Protection Devices						
Part Number	Price	QTY	Description	Nominal Input Voltage	Dimensions & Package	Mating Socket
AD-ASMD-250	\$12.50	5	Protection diode module for 783, 784 and 75 series relays.	6-250 VDC	Figure 1	783-3C-SKT 784-4C-SKT-1 750-2C-SKT 750-3C-SKT
AD-ASMM-24	\$10.50	5	MOV module for 783, 784 and 75 series relays that operate at 24 VAC coil voltage.	24 VAC/VDC		
AD-ASMM-120	\$10.50	5	MOV module for 783, 784 and 75 series relays that operate at 120 VAC coil voltage.	120 VAC/VDC		
AD-ASMM-240	\$10.50	5	MOV module for 783, 784 and 75 series relays that operate at 240 VAC coil voltage.	240 VAC/VDC		
AD-BSMD-250	\$10.50	5	Protection diode module for 782 series relays.	6-250 VDC	Figure 2	782-2C-SKT
AD-BSMM-24	\$10.50	5	MOV module for 782 series relays that operate at 24 VAC coil voltage.	24 VAC/VDC		
AD-BSMM-120	\$10.50	5	MOV module for 782 series relays that operate at 120 VAC coil voltage.	120 VAC/VDC		
AD-BSMM-240	\$10.50	5	MOV module for 782 series relays that operate at 240 VAC coil voltage.	240 VAC/VDC		

Dimensions

inches [mm]

Figure 1

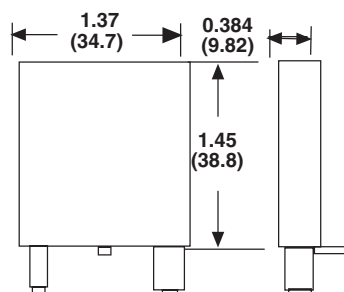


Figure 2

